

Tran Thi Thao Nguyen

AI Intern

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GitHub: github.com/nguyentran206 **Portfolio:** nguyentran-portfolio.vercel.app

Summary

Computer Science student with hands-on experience in NLP and Machine Learning projects, including machine translation and toxic text classification. Interested in LLMs, NLP, and applied AI systems, with practical experience in model fine-tuning, evaluation, and data preprocessing.

Education

Ton Duc Thang University

Ho Chi Minh City, Vietnam

Bachelor of Science, Computer Science. GPA: 7.83/10 (Current)

Expected Graduation: 2026

Relevant Coursework: Data Structures, Introduction to Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing, Massive Data Processing.

Experience

Titan Technology Corporation

Ho Chi Minh City, Vietnam

Development & Business Analyst Intern

March 2026 – May 2026

- Analyzed business and system requirements, reviewed technical documentation, and identified inconsistencies across specifications, process flows, and data models.
- Developed and tested backend APIs using Python, SQLAlchemy, PostgreSQL, and Postman while supporting AI-assisted business analysis workflows.

Projects

Machine Translation Model

GitHub: Machine-Translation

- Fine-tuned model ViT5-base for Vietnamese to English machine translation with cleaned bilingual data and GPT-generated synthetic pairs. Applied a DPO-inspired preference optimization using GPT-evaluated chosen/rejected outputs to enhance translation fluency.
- Built an end-to-end training pipeline including tokenization, data augmentation, fine-tuning, and evaluation with BLEU, chrF and ROUGE metrics, complemented by LLM-based qualitative assessment using Gemini.
- Analyzed performance differences across model families and identified trade-offs in generalization, sequence fluency, and alignment with human-like translations.
- Achieved improved translation quality with BLEU (38.853), chrF (60.692) and ROUGE-L (0.623) scores after fine-tuning, supported by Gemini-based qualitative evaluation (average score: 7.3/10) on fluency, adequacy and faithfulness.

Vietnamese Toxic Text Classification

GitHub: Vietnamese-Toxic-Text-Classification

- Developed a Vietnamese toxic comment classification pipeline using TF-IDF and PhoBERT embeddings with Logistic Regression, Naive Bayes, and SVM classifiers.
- Performed data cleaning, text normalization, and feature engineering, and systematically compared traditional ML models with embedding-based approaches.
- Achieved 86.9% accuracy on the ViHSD & Kaggle dataset, with detailed error analysis across different types of toxic language.

Skills

Programing & data: Python, SQL, Pandas, Numpy.

Machine Learning & Deep Learning Frameworks: TensorFlow, PyTorch, Scikit-learn.

Tools: Git/GitHub, Visual Studio Code, Google Colab, Kaggle, VastAI.

Laboratory: Data Preprocessing, Model Training, Evaluation Metrics.

Language: Vietnamese (Native), English (B2 – CEFR, Aptis Certified).